PROJECT PLAN

STARLIGHT

Starbucks Treatment And Location Inference for Gentrification & Housing Trajectory

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Thesis

The widely-cited claim that homes near Starbucks appreciate 96% versus the 65% national average conflates two distinct effects: **(1)** Starbucks's skill at selecting neighborhoods already on an appreciation trajectory, and **(2)** the causal effect of a new Starbucks on local home prices. This project decomposes the headline number into its selection and causal components using Atlanta as a testbed, builds a doubly-robust difference-in-differences estimator that handles both confounders, and deploys the resulting models as a public API that, given any Atlanta address, returns expected appreciation properly accounting for the probability of a Starbucks opening and the genuine causal impact if one does.

Why Atlanta

- **BeltLine natural experiment.** The 22-mile trail loop opened in staggered segments starting around 2008. Cleanest urban-gentrification natural experiment in any US city, and it coincides with Starbucks's recent expansion.
- **Interpretable heterogeneity.** East-west divide, intown vs OTP, and varied gentrification trajectories give findings a clear narrative.
- **Good open data.** Fulton County, DeKalb County, Atlanta Open Data, MARTA, and the BeltLine all publish usable datasets.
- **Dramatic Starbucks variation.** Long-standing stores in Buckhead and Midtown; recent expansion into West End, Reynoldstown, Edgewood, and East Atlanta Village provides high treatment-timing variance.

High-Level Plan: Four Phases

Phase	Weeks	Goal
1. Data foundation	1–2	Clean tract × month panel with prices, treatment status, and covariates.
2. Site-selection model	3–4	Calibrated classifier predicting P(Starbucks opens) with SHAP explanations.

3. Causal model	5–6	Doubly-robust DiD producing naive, controlled, and DR treatment effects.
4. Deployment	7–8	FastAPI live, simple map frontend, README with the narrative.

Phase 1 — Data Foundation

Goal: a clean panel dataset of (tract × month) with home prices, Starbucks treatment status, and all covariates ready to model.

Geographic unit

Use **census tracts** as the primary panel unit. Small enough that *a Starbucks opened here* is meaningfully local, big enough for stable monthly price estimates, matches ACS natively, and is the standard unit for staggered DiD in urban econ. The site-selection model in Phase 2 uses a finer unit (H3 hexagons at resolution 9, ~150m across).

Datasets

Starbucks locations and opening dates (the hard one)

- **Current locations:** Starbucks store-locator JSON endpoint. ~150 stores in metro Atlanta.
- **Opening dates** (not published — triangulate from three sources):
- **Wayback Machine + store locator:** scrape snapshots of starbucks.com/store-locator at 6-month intervals from 2010–2026; diff snapshots to find new store numbers.
- **Fulton/DeKalb business license records:** filter for Starbucks filings.
- **Google Places first review date:** lower bound on opening — first review is usually within 30 days.
- **Validation:** cross-reference all three sources; manually verify ~10% by Google News.

Output: `starbucks_openings.csv` with `store_number`, `lat`, `lon`, `tract_geoid`, `opening_year_month`, `source`, `confidence`.

Home prices

- **Primary panel:** Zillow ZHVI at census-tract level, monthly, 2000–present. Free CSV.
- **Validation / robustness:** Fulton and DeKalb tax assessor parcel-level transaction records.

Demographics and time-varying controls

- **ACS 5-year estimates** at tract level, 2010–2024: median income, education, race/ethnicity, age, household size, % renter, median rent.
- **LEHD LODES:** tract-level daytime population and worker inflow/outflow — critical for the site-selection model.
- **BLS QCEW:** county-level employment by industry as a metro-wide time control.

Built environment

- **OpenStreetMap:** coffee shops, restaurants, gyms, banks, grocery, schools, parks (Overpass API or Geofabrik dumps).

- **BeltLine trail openings** by segment, with dates.
- **MARTA** stations and lines (public GTFS).
- **Atlanta zoning shapefiles** for filtering eligible Starbucks sites.

Other anchor retail (confounders)

Whole Foods, Trader Joe's, Sweetgreen, Chipotle, Target — current locations as covariates; opening dates only for brands you specifically control as time-varying confounders.

Panel structure

```
tract_geoid | year_month | zhvi | treated | months_since_treatment
            | acs_median_income | acs_pct_college | daytime_pop
            | n_coffee_shops_quarter_mi | distance_to_beltline | ...
```

One row per (tract, month). ~500 tracts × ~150 months ≈ **75K rows**. Manageable in pandas. `treated = 1` for tract-months at or after the first Starbucks opened in that tract; never-treated tracts are pure controls. Staggered-DiD methods handle this structure natively.

Phase 1 deliverable

A reproducible pipeline (Python scripts + Makefile, or Prefect/Dagster) that builds `panel.parquet` from raw sources, plus a notebook with basic EDA: tracts treated, treatment-timing distribution, parallel-trends plot of raw ZHVI.

Phase 2 — Site Selection Model

Case-control sampling

- **Positives:** every Starbucks opening location (~150 from Phase 1).
- **Negatives:** 5–10× as many comparable, retail-eligible parcels (commercially zoned, min parcel size, accessible).
- **Temporal alignment:** for each positive opened in year T, use features as of T–1; for negatives, sample random T and use T–1 features. Prevents post-treatment leakage.

Features

Built from the panel, sampled at H3 res-9 around the candidate site:

- **Catchment demographics** (ACS within 0.25, 0.5, 1.0 mi rings): income, education, age, race, household type
- **Daytime population** (LEHD): workers per acre within 0.25 mi
- **Commuter flow** (LEHD): inflow / outflow ratio
- **Co-tenancy** (OSM + retail): count of each anchor brand within 0.25 mi
- **Competitor density:** existing coffee shops within 0.1 and 0.25 mi
- **Cannibalization:** distance to nearest existing Starbucks
- **Accessibility:** nearest road class, distance to nearest signalized intersection, MARTA proximity, BeltLine proximity
- **Visibility proxies:** corner lot (boolean), parcel frontage
- **Real estate:** parcel size, building sqft (county assessor)

The model

LightGBM binary classifier, hyperparameters via Optuna with **grouped CV** (group by tract to prevent leakage of Starbucks's clustering pattern). Calibrate probabilities with isotonic regression on a held-out set so the output is a real propensity score, not a ranking. SHAP for explanations.

Validation: PR-AUC (class-imbalanced), Brier score (calibration), and a **temporal holdout** — train on openings before 2022, test on 2022–2025. Report: *of my top 100 predicted opportunity sites in 2022, X% became actual Starbucks by 2025.*

Phase 2 deliverable

Trained model artifact `site_selection_v1.lgb` + notebook with SHAP plots + a `score_location(lat, lon)` function returning a calibrated P(Starbucks opens) + temporal-holdout numbers.

Phase 3 — Causal Model

This is where the rigor lives. Three nested estimates, each strictly more credible than the last.

Estimate 1: Naive DiD

Standard two-way fixed effects on the panel:

```
log(zhvi) ~ treated + tract_FE + month_FE
```

Report the coefficient on `treated` as the naive effect. This will be inflated — *that's the point*. It is the baseline you are going to beat.

Estimate 2: Staggered DiD (Callaway–Sant'Anna)

Use the `differences` Python package (or call R's `did` via rpy2). CS handles staggered treatment timing without the bias that pure two-way FE suffers when effects vary.

Report:

- Overall ATT (average treatment effect on the treated)
- Event-study plot: effect by months-since-treatment; pre-period = parallel-trends test
- Heterogeneous effects by cohort (tracts treated in 2018 vs 2022, etc.)

Estimate 3: Doubly-Robust DiD

This is the headline result. Combine:

- The site-selection model (Phase 2) as the **propensity score** for treatment
- A regression model for the outcome (log price) given covariates

The DR estimator (Sant'Anna–Zhao 2020) reweights treated and control units by the inverse of the site-selection propensity, then runs DiD on the reweighted sample. **Key property:** if *either* the propensity model or the outcome model is correct, the estimate is consistent.

Headline finding

```
selection_bias_share = (estimate_2 - estimate_3) / estimate_2
```

If this is, say, 0.6, the README says: *60% of the apparent Starbucks-appreciation premium reflects Starbucks's site-selection skill, not a causal effect of new stores on prices.*

Phase 3 deliverable

Notebook with all three estimates, event-study plots, parallel-trends robustness checks, the placebo test (announced-but-not-opened locations), and a plain-language interpretation. Saved estimates feed the FastAPI `/predict_impact` endpoint.

Phase 4 — Deployment

Stack

- **FastAPI** for the API
- **Pydantic** for request/response schemas
- **Uvicorn** for the server
- **Render** or **Fly.io** for hosting (free tiers, both fine for FastAPI)
- **Leaflet + single-page HTML/JS** served as a static file from the same FastAPI app
- **OpenStreetMap or Mapbox tiles** for the basemap
- Skip Docker for V1 unless already comfortable — `requirements.txt` + Render config is enough.

Endpoints

```
POST /score_location        -> site-selection score + SHAP top features
POST /predict_impact        -> stacked: P(opens) x causal effect
GET  /opportunity_zones     -> top-N hexes ranked by expected impact
GET  /tract_summary/{geoid} -> all data for one tract (map detail view)
```

Example /predict_impact response

```
{
  "starbucks_viability_score": 0.78,
  "p_starbucks_opens_within_5yr": 0.34,
  "predicted_appreciation_if_opens": {
    "naive_did_estimate": 8.4,
    "doubly_robust_estimate": 3.1,
    "selection_bias_share": 0.63
  },
  "expected_appreciation": 1.05,
  "interpretation": "Moderate viability, but selection bias means
                     the headline number overstates the true causal
                     effect by ~63%."
}
```

`expected_appreciation = p_opens × causal_effect` — integrating over whether Starbucks actually opens. A quantity neither sub-model produces alone.

Map demo

Single-page Leaflet map of Atlanta. Tracts colored by expected appreciation. Click a tract → side panel shows the decomposition. Toggle existing-Starbucks layer. **This is your portfolio screenshot.**

Phase 4 deliverable

Live URL · README with the narrative structure · reproducible repo · 5-minute Loom walkthrough (optional, high ROI).

README Narrative Structure

The story flow that hiring managers will remember:

1. The hook. Everyone has seen the 96% vs 65% appreciation stat. What's actually going on?

2. The naive answer. Basic correlation. Confirms the stat.

3. First refinement. DiD around store openings. Effect shrinks but stays large.

4. The selection problem. Starbucks does not open at random. Train a model on their historical site selection to learn *how* they pick.

5. Findings from the site model. What Starbucks actually weights — daytime population, co-tenancy with X, commuter flow — with SHAP plots.

6. The doubly-robust answer. Combining the propensity with DiD gives a much smaller causal effect. The headline overstates by ~Xx.

7. The product. Deployed as a tool that, given any address, returns expected appreciation properly accounting for both whether Starbucks would open and how much it would actually move prices.

8. The bonus. Opportunity-zone visualization showing high-impact under-served locations.

Repo Structure

```
starlight/
■■■ README.md
■■■ pyproject.toml           # or requirements.txt
■■■ Makefile                 # for reproducibility
■■■ data/
■   ■■■ raw/                 # gitignored, populated by scripts
■   ■■■ interim/
■   ■■■ processed/
■       ■■■ panel.parquet
■■■ src/
■   ■■■ data/
■   ■   ■■■ starbucks_scraper.py
■   ■   ■■■ wayback_scraper.py
■   ■   ■■■ census_pull.py
■   ■   ■■■ osm_pull.py
■   ■   ■■■ build_panel.py
■   ■■■ models/
■   ■   ■■■ site_selection.py
■   ■   ■■■ causal_did.py
■   ■   ■■■ stacked.py
■   ■■■ api/
■   ■   ■■■ main.py          # FastAPI app
■   ■   ■■■ schemas.py       # Pydantic models
■   ■   ■■■ inference.py     # model loading + prediction logic
■   ■■■ frontend/
■       ■■■ index.html       # Leaflet map
■■■ notebooks/
■   ■■■ 01_eda.ipynb
```

```
■   ■■■ 02_site_selection.ipynb
■   ■■■ 03_causal_naive.ipynb
■   ■■■ 04_causal_did.ipynb
■   ■■■ 05_causal_doubly_robust.ipynb
■   ■■■ 06_validation.ipynb
■■■ models/                  # serialized .lgb, .pkl artifacts
■■■ tests/
    ■■■ test_api.py
```

Tools / Libraries

Python 3.11+ · pandas · geopandas · h3-py · lightgbm · optuna · shap · statsmodels · **differences** (Callaway–Sant'Anna and DR-DiD in Python — the key library for Phase 3) · scikit-learn · fastapi · pydantic · uvicorn · httpx (scraping). Visualization: matplotlib · plotly. Frontend: Leaflet + vanilla JS (no React needed).

Realistic Risks and Mitigations

Risk: Wayback Machine scraping for opening dates is brittle.

Mitigation: Have the county business-license fallback ready from week 1; if Wayback is noisy, lean on that as the primary.

Risk: Parallel trends fails for many treated tracts.

Mitigation: This is a *finding*, not a failure. It means selection bias is real and is exactly why the doubly-robust estimator matters. Lean into it in the README.

Risk: DR-adjusted causal effect is close to zero or negative.

Mitigation: Also a valid finding. Headline becomes *the Starbucks effect is mostly selection* rather than *Starbucks causes X% appreciation*. Either result is publishable-quality.

Risk: Scope creep on site-selection features.

Mitigation: Set a feature-set budget in week 3 and stop adding once you hit it.

This Week

- **Day 1–2:** set up the repo. Get the Starbucks store-locator scraper working for current Atlanta locations. Confirm Census ACS API access. Pull Atlanta tract shapefiles.
- **Day 3–4:** start the Wayback Machine scraper for historical Starbucks snapshots. Run it in the background — one snapshot per quarter from 2010 to present.

- **Day 5–7:** download ZHVI tract-level data; basic EDA — how many Atlanta tracts ever got a Starbucks, what does the raw appreciation gap look like, plot on a map.

That EDA at the end of week 1 is your decision point. If Atlanta's tract-level data clearly shows the appreciation gap, proceed with confidence. If it doesn't, reconsider city choice before sinking more time.

Project plan generated for an undergraduate ML / informatics portfolio piece. Atlanta scope V1 → Houston/national replication V2.